SOP-PROJECT MANAGEMENT IN INDUSTRIAL SOLUTIONS

[CONFIDENTIAL]
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CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Date Prepared By	Date Reviewed By	Date Approved By
1	0.1	Draft	Implementation of the Integrated Management System	Aniket Parab 29-Feb-24		
2	Issue B Version 1.0	Revised the Document to suit the requirements for implementation of the integrated management system			Ramya T V 12-Mar-24	
3	Issue B Version 1.1	Revised Section 8a, 8b, 8c, process flow for deallocation of resources deleted. RTM replaced with Functional Design Specifications/control philosophy. FAT/testing-Internal or external is added before the dispatch of material.	Review comments	Aniket Parab 14-Mar-24	Sanjeeva G Byakod 22-Mar-24	
4	Issue B Version 1.2	Addition of Metrics-CSAT & Margin at Closure	Review comments	Aniket Parab 25-Mar-24	Sanjeeva G Byakod 12-Mar-2024	
5	Issue B Version 1.3	No changes	Approval			Sanjeeva G Byakod 12-Mar-2024

6	Issue B Version 1.4	Document formatting done as per document guidelines	Updated in line with the document guidelines	Obul Prasad Reddy 20-Dec-24	Anitha N 23-Dec-2024	Santosh V 31-Dec-2024
7	Issue C Version 1.0	Review of document	New version update of IMS	Anitha N 01-Aug-2025	Santosh V 08-Aug-25	Sanmangala M S 14-Aug-2025

1 Purpose

To establish a standardized procedure for planning, managing, executing, and closing projects within the Industrial Solutions division at Utthunga.

2 Scope

This applies to all the Industrial Solution projects executed in Utthunga.

3 Abbreviations

a. PMO: Project Management Officer

b. SME: Subject Matter Expert

c. PE: Process Excellence

d. IT: Information Technology

e. PP: Project Plan

f. SOW: Statement of Work

g. PO: Purchase Order

h. PM: Project Manager

i. AHF: Activity Handover Form

j. FDS: Functional Design Specifications

k. CRS: Customer Requirement Specifications

l. UT: Utthunga

m. SAT: Site Acceptance Test

n. RS: Requirement Specifications

o. ISG: Infrastructure Service Group

p. SA: System Administrator

q. CSAT: Customer Satisfaction

r. FAT: Factory Acceptance Test

s. MOM: Minutes of Meeting

4 Terms and Definitions

NA

5 Responsibilities

a. PM shall be responsible for initiating the project with internal & external kick off meetings.

b. PMO shall allocate the resource required for the project.

c. PM shall prepare the PP, control budget and execute the project as per PP and other relevant project documents as listed in this SOP.

d. PM shall manage the project and ensure that delivery is done on time and meeting quality objectives.

e. PM shall lead the team to identify the risks associated with the project.

f. PM along with team members shall identify the mitigation and contingency plans for the risks identified.

g. PM shall ensure that the mitigation plans are implemented, and their effectiveness is recorded.

h. PM shall communicate to on site/offshore team members.

i. PM or person designated by PM shall communicate with customer.

j. PM shall coordinate and direct the subcontractors.

k. BU head/Operation Director and PM shall review the work progress weekly.

l. PM and PE owner shall ensure the scope of work required for the delivery is completed.

m. PM and Project Team Members shall ensure customer satisfaction.

n. PE Owner to Collect Customer feedback and measure CSAT score.

o. SA shall terminate all access requirements post project completion.

6 RACI

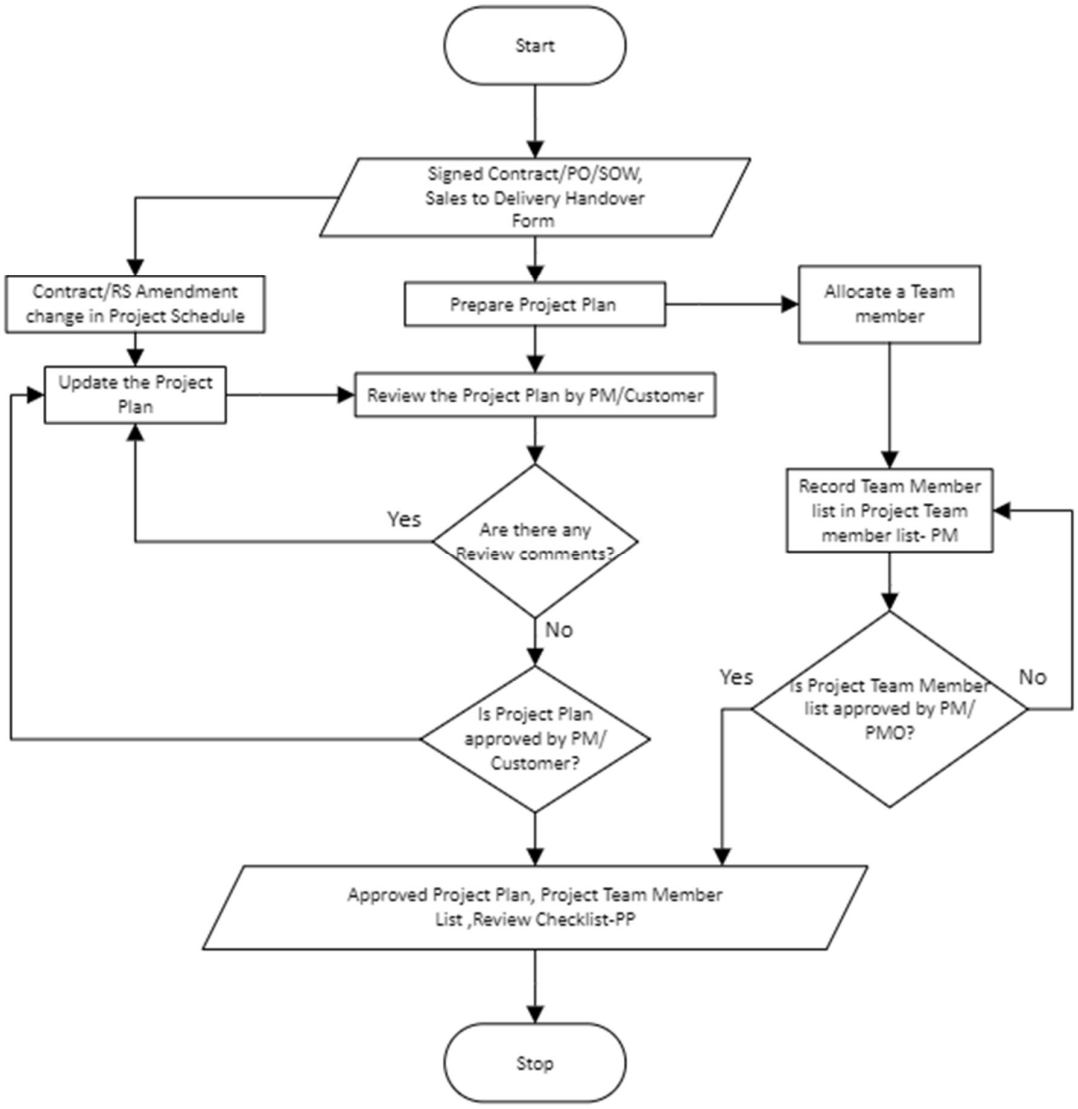
Process	Responsible	Accountable	Consulted	Informed
Project Planning	Project Manager	Project Manager	BU Head, PMO, Operations Head	Customer/ PE Team/ Accounts
Project Execution	Project Team Members	Project Manager	SME's, PE Team, Operations Head, BU Head	Customer
Project Handover	Project Team Members	Project Manager	Customer, Delivery Team, PE team, BU Head	Customer/ PE/ IT / Accounts/ Sales

7 SIPOC

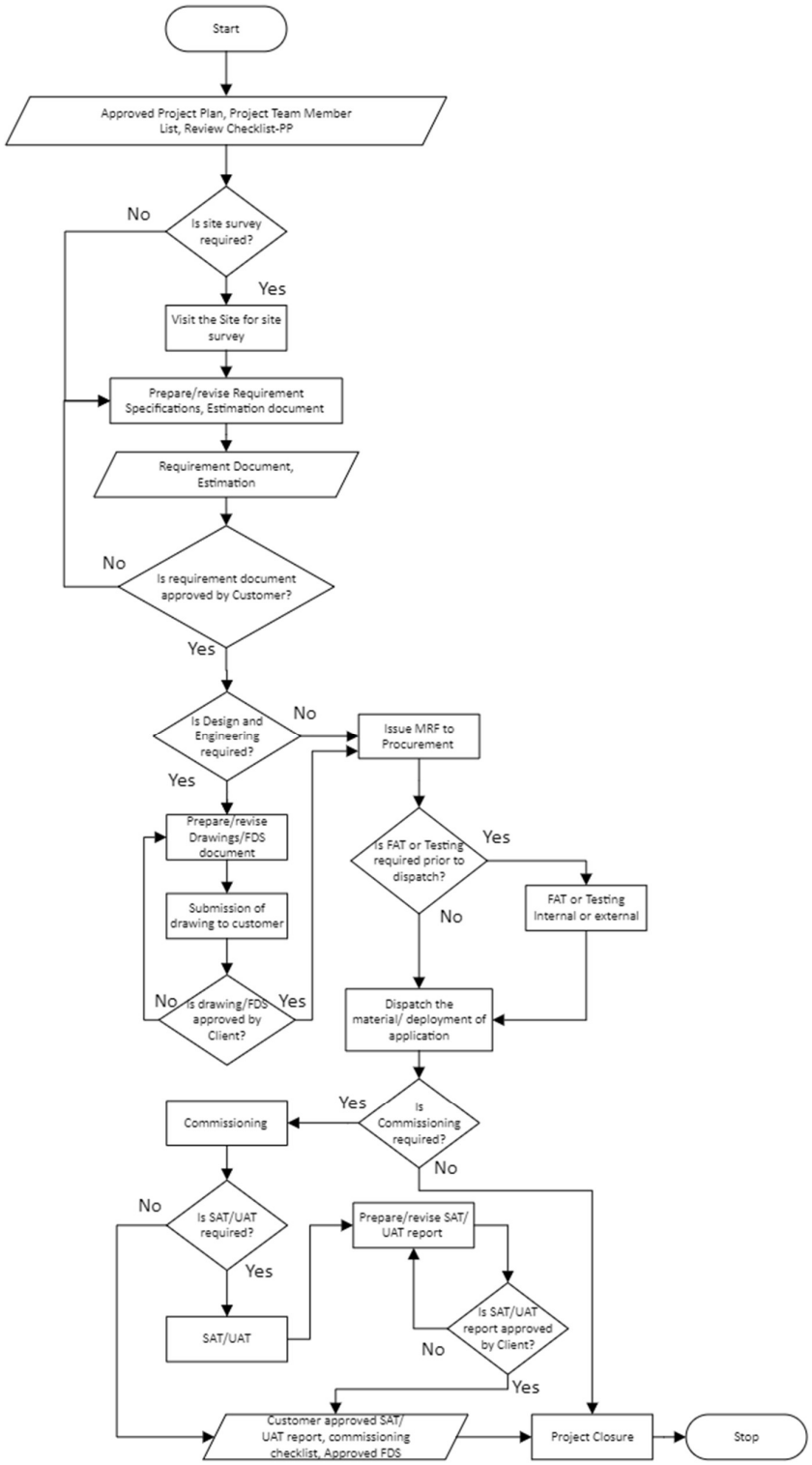
Supplier	Input	Process and Control		Output	Customer
		Process	Control		
Project Manager	Acknowledged Signed Contract/PO, Sales to delivery handover form, historical data from past projects/ lessons learnt	Project Planning	Project plan workbook, reviews	Approved Project Plan	Project Team, Customer
Project Team	Approved Project Plan, Requirement specifications	Project Execution	Risk Management, Project Plan, reviews, change control	Deliverables including but not limited to Test certificates, Commissioning reports etc.	Customer
Project Manager	Acceptance of Deliverables by customer	Project Closure	Project closure checklist, workbook	Project Delivery, Resource release, Contract closure and Archival of project	Customer

8 Overall Process Flow Chart

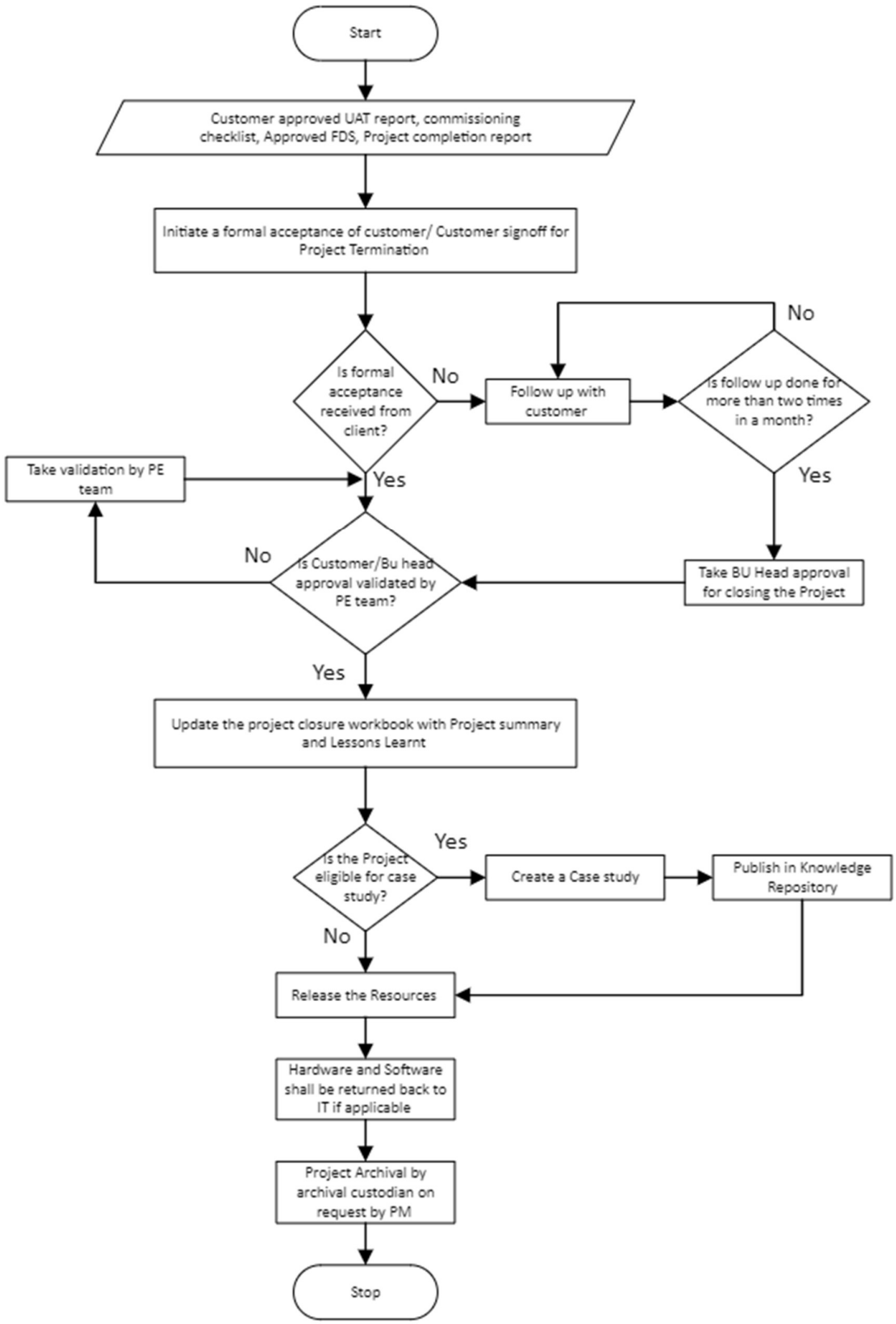
a. Project Initiation



b. Project Execution



c. Project Closure



9 Process Description

This section outlines the key steps involved in managing projects within the Industrial Solutions Division. The process described here is specifically applicable to the Automation/Digitalization (Ax/Dx) and Process Engineering sub-divisions. However, note that separate SOPs exist for the other sub-divisions of Industrial Solutions.

9.1 Project Planning

a. The Salesperson will coordinate the project handover to the engineering team via the BU Head. The BU Head will nominate the Project Manager.

b. The Project Manager shall ensure that on receipt of the signed contract and/or project initiation email or letter of intent from sales, project planning shall be done based on the reviewed requirements for the project.

c. In absence of Signed Contract, Project can be initiated with email confirmation from Client and Utthunga' s Sales team. Validity of mail intent order can be extended maximum by 1 month. In case of non-extension of the order, BU head has to approve, failing which resources allocated to project will be de-tagged and same will be communicated to customer.

d. PM will arrange an internal kick off meeting with all the required stakeholders as per project requirement like HR, Accounts, and Quality, and Procurement.

e. During the project's internal kick-off meeting, PM must utilize the Project Kick-off Checklist.

f. Project schedule, Budget and resource allocation will be decided during this internal kick off meeting.

g. PM shall arrange for an external kick off meeting with the customer for all project excluding the supply only projects unless the customer requests otherwise.

h. For Supply type projects, the necessity of an external kick off meeting will be at the discretion of the PM, who may opt to conduct one if deemed necessary prior to project commencement or based on specific customer requirements.

i. For external kick off meetings Project Kickoff Deck-External PPT template shall be used.

j. After the completion of kick off meeting PM shall record the minutes of meeting as per MOM template and shall share the MOM to the customer and the project team.

k. Project Manager shall ensure that commitments for the reviewed requirements / changes to reviewed requirements are discussed and agreed upon with the affected groups.

l. The project manager, in conjunction with team members, will perform a Risk Assessment to identify potential risks, develop mitigation strategies, and establish contingency plans.

m. For supply-type projects risk assessment shall be conducted selectively based on the size and complexity of the project. This determination will be made by the project manager in consultation with Delivery Manager.

n. Risk assessment for supply projects not subject to selective assessment will be based on information from risk register available in project repository, ensuring comprehensive coverage of potential risks.

o. The risk planning shall be done based on the Risk Management Guidelines and shall be recorded in Project Risk Management Template.

p. Risk assessments for projects is required to be conducted periodically and the frequency for conducting the risk assessment shall be as defined in the Project Plan.

q. Project plan document shall be prepared as per the Project Plan Template.

r. Project plan shall be appropriately updated, when there are changes to the reviewed requirements and based on the Project / Phase review action items, Internal Quality Audits.

s. Changes to the project plan shall be reviewed and agreed upon with the stakeholders appropriately.

t. Review and approval of Project Plan shall be done as described in the Review Process.

u. Project Plan Review Checklist shall be used in the review of the project plan.

v. Review findings shall be recorded in the Review Form and tracked to closure as described in the Review Process.

w. Refer Review process for further details.

x. PM shall use the PAL to refer to similar projects estimation sheets, Risk Database, Measurement Database to plan for the current project.

y. PM shall compare the required and available skillset of the project resources to execute the project and document the gap using the project training plan.

z. Identify the training requirements and communicate to L&D.

aa. Identify all the relevant stakeholders and define respective roles and responsibility along with communication mechanism.

bb. Check if re-planning is required or not based on the monitoring result.

cc. Re-planning should be carried out for the following changes (but not limited to) and approval must be taken from BU head & Customer before execution.

- Resource Change

- Scope Change
- Technology Change
- Infrastructure Change
- Critical project milestone
- Any other external triggers

dd. The details of members working for a project, including their competency level, shall be recorded in the Project Team Members List.

ee. When a project member is added or deleted from the project, the project team members list shall be appropriately updated.

ff. Any team member leaving the team/ transferred to another project/ going onsite will do a formal handover of activities to a person identified by the Project Manager. This shall be recorded in the Activity Handover Form.

gg. Project Manager shall approve this activity handover form and shall update the project team member list.

9.2 Project Execution

a. Once the PP is approved by PM or customer, site survey will be initiated if required as per Contract/PO.

b. The requirement specification document and estimation document shall be prepared by the person designated by PM.

c. Any change requests arising out of changes in the following, but not limited to, scope, design, process, material, notification, and communication, shall be managed as defined in the Engineering Change Request Management SOP.

d. BU head shall approve the estimation document after the review by PM, if any changes the estimation document needs to be revised accordingly.

e. PM shall send the requirement specification to customer for approval.

f. The team will also work with the customer contact for any clarification and early feedback during the preparation.

g. After the approval is received from customer, MRF will be raised to Procurement team. Procurement of all the materials shall be carried out as per procurement SOP.

h. Procurement team shall provide the vendor documentations related to procurement as per MRF.

i. Technical validation of vendor documents or materials shall be done by the project technical team.

j. Project execution will be based on the type of projects:

- Design Engineering and Development

- The engineering Team shall prepare the drawings, FDS documents required as per the requirement specifications/PO.
- PM or person designated by PM shall share the drawing/FDS to customer for review & approval.
- Once the approval is received from customer, PM shall provide dispatch clearance to procurement team.
- After the material is received, it shall be dispatched to customer site.
- In some cases, if customer request for Factory acceptance test (FAT), the system will be shipped after FAT is approved by customer.

- Supply
  - For supply items PM shall provide dispatch clearance to procurement team.
  - Once the material is received it shall be dispatched to customer site.
  - In some cases, if customer request for Factory acceptance test (FAT), the system will be shipped after FAT is approved by customer.

k. Once the material is received at site, based on the customer confirmation on the proposed schedule, the team shall visit the site to install and commission the system.

l. After the commissioning the team shall perform the site acceptance test (SAT)/ User Acceptance Test (UAT) if applicable as per the project requirement stated in FDS/SOW/PO.

m. Post completion of SAT/UAT, team shall prepare the SAT/UAT report using the customer provided format or in Utthunga template for SAT/UAT report and share with customer for approval.

n. Team shall train the users in the end customer site and handover the system.

9.3 Project Closure

a. Project Manager is responsible to close the project contract.

b. Project Manager or person designated by PM is responsible for getting the formal acceptance/ completion certificate from customer.

c. If formal acceptance is not received after two follow-ups within one month of project completion, the project may be closed with approval from the BU head.

d. In case of force termination of a project, Head-Sales and Marketing, Function Head, Legal department shall review the contract and legal documents for its impact and justification shall be provided for the same. Root Cause Analysis shall be performed, and the report shall be stored as part of project closure records.

e. All the project details like effort variance, schedule variance, risks encountered, best practices should be clearly documented in the Project Summary form.

f. Lessons learnt should be documented in Lessons_Learnt_Report and the same should be submitted to Quality Department.

g. Delivery manager along with quality team shall determine of whether the project is qualified for a case study and will assess it based on the criteria outlined below.

- Project that faced technical, organizational, or business challenges that make it interesting to study and learn from
- Project that met Organizational goals and delivered exceptionally.
- Projects which aim to address broader issues or contexts from different domains, industries, or technologies, exceptional project management which can enhance its applicability if made a case study.
- Project that is practical to study given the available resources, such as time, budget, and access to information.
- The project's scope and Size. It may be a large-scale project involving multiple teams or a smaller project with specific characteristics.

h. Case study shall be created without disclosing any customer details or confidential details and will be published in knowledge repository.

i. The Hardware and Software shall be returned to the IT Support department for further actions.

j. PM shall inform the PMO to release the team members and shall intimate to Resource Management about their updated skill sets.

k. Data archiving activities shall be performed as defined in Backup, Security and Archival Process (refer section 9.1.14 Archival).

l. PM shall be responsible to get a confirmation to send or retain customer supplied material.

10 Information Security requirements

a. Secure access (deregistration) to various applications, Project portals.

b. Ensure data privacy, confidentiality and IP requirements are clearly specific in the contract.

c. Information Security requirements in the contract are clearly put into place.

d. Compliance requirements based on different geographies are adhered to projects.

e. Confidentiality of information and project details is maintained by the team.

f. Archiving and back up of data as per contractual obligations.

11 References

a. ISO 9001 Clauses 8.1, 8.2, 8.3, 8.5, 8.6, 8.7

b. ISO 27001 Clauses 8.1, Controls A.6, A.9, A.10, A.14

c. ISMS Control Policies

d. Risk Management Guidelines UT/PM/PPP/GDL/01

e. Customer Feedback (CFP)-UT/PS/CFP/SOP

f. Review Process (RVP)-UT/PS/RVP/SOP

g. Configuration Management (CMP)-UT/PS/CMP/SOP

h. Requirements Specification (RSP)-UT/SD/RSP/SOP

i. Measurement, Analysis, and Improvement (MAI)-UT/IM/MAI/SOP

j. Externally Provided Property-UT/IM/EPP/SOP

k. Engineering Procurement Construction and Commissioning SOP -

12 Metrics

a. Online Milestone completion time Delivery

b. Project Completion (Schedule Variance)

c. Ontime Material receipt to site.

d. Effort Variance

e. No. of Quality nonconformance

f. Requirement stability index

g. Resource stability index

h. Number of recovery plans (revisions to project plan)

i. Budget VS Actual

j. CSAT

13 Annexures

a. NA

14 Forms

a. Activity Handover Form UT/PM/PPP/TMP/02

b. Project Plan Workbook UT/PM/PPP/TMP/04

c. Project Plan Review Checklist UT/PM/PPP/TMP/07

d. Risk Identification Checklist UT/PM/PPP/TMP/08

e. Project Kick Off Checklist UT/PM/PPP/TMP/12

f. Project Kickoff Deck-External

g. Project Weekly Status Report UT/PM/PPP/TMP/16

h. Project ID creation Request form UT/PM/PPP/TMP/21

i. Project End Date Extension Form UT/PM/PPP/TMP/28